MANIFOLDS

DEF: a topological space M is called an n-dimensional **TOPOLOGICAL MANIFOLD** if:

1. M is locally homeomorphic to $\mathbb{R}^n$, i.e. $\forall p \in M \; \exists U \subset M$ open, $p \in U \; \exists V \subset \mathbb{R}^n \; \exists$ homeom. $x: U \to V$;
2. M is Hausdorff, i.e. $\forall p \neq q \in M, \exists U_1, U_2 \subset M$ open $p \in U_1, q \in U_2$, $U_1 \cap U_2 = \emptyset$;
3. The topology of M has a countable basis, i.e. $\exists U_1, U_2, \ldots \subset M$ countably many opens in M s.t. each $U \subset M$ open can be written as $U = \bigcup_j U_j$.

TH. 1: Let M and A be sets. $\forall \alpha \in A$ are given subsets $U_\alpha \subset M$ and $V_\alpha \subset \mathbb{R}^n$ and bijective maps $x_\alpha: U_\alpha \to V_\alpha$. Suppose:

1. $M = \bigcup_{\alpha \in A} U_\alpha$
2. $x_\alpha(U_\alpha \cap U_\beta) \subset V_\alpha$ open in $\mathbb{R}^n$ $\forall \alpha, \beta \in A$ ($\overset{\alpha=\beta}{\Rightarrow}$ V_α open)
3. $x_\beta \circ x_\alpha^{-1}: \underset{\subset \mathbb{R}^n}{x_\alpha(U_\alpha \cap U_\beta)} \to \underset{\subset \mathbb{R}^n}{x_\beta(U_\alpha \cap U_\beta)}$ continuous $\forall \alpha, \beta \in A$.

Then, $\exists!$ topology on M s.t. $U_\alpha \subset M$ open and $x_\alpha: U_\alpha \to V_\alpha$ are homeomorphism $\forall \alpha \in A$.

EX: $\mathbb{RP}^n = \mathbb{P}(\mathbb{R}^{n+1}) = \{L \subset \mathbb{R}^{n+1} \mid L \text{ 1-dim. vector subspace}\}$

$M = \mathbb{RP}^n$, $A = \{\text{affine-linear embedding } \alpha : \mathbb{R}^n \hookrightarrow \mathbb{R}^{n+1} \text{ s.t. } 0 \notin \alpha(\mathbb{R}^n)\}$

$\alpha \in A$, $\exists\, B \in M_{(n+1)\times n}(\mathbb{R})$ of rank n (α embedding),
$\exists\, c \in \mathbb{R}^{n+1}$ s.t.

$$\alpha(x) = Bx + c, \quad \alpha(\mathbb{R}^n) = \text{hyperplane}.$$

$U_\alpha = \{L \in \mathbb{RP}^n \mid L \cap \alpha(\mathbb{R}^n) \neq \emptyset\}$ (I'm excluding lines parallel to $\alpha(\mathbb{R}^n)$)

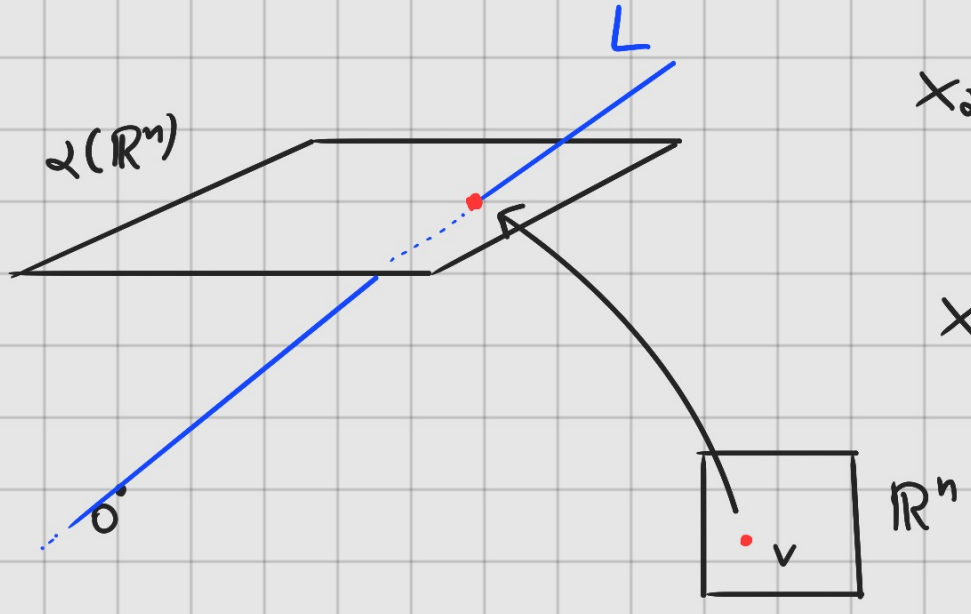
if they intersect, they intersect in exactly 1 point.

$V_\alpha = \mathbb{R}^n$

$x_\alpha : U_\alpha \to V_\alpha$

$x_\alpha(L) = \alpha^{-1}(L \cap \alpha(\mathbb{R}^n))$
(α bijective on image).

$x_\alpha^{-1}(V) = \mathbb{R} \cdot \alpha(v)$

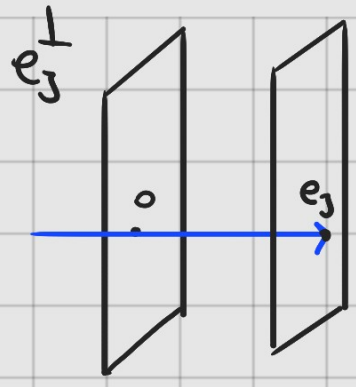


$e_0, \dots, e_n$ std ONB of $\mathbb{R}^{n+1}$

(1) For $j = 0, \dots, n$,

$v \in \mathbb{R}^n$, $\alpha_j(v) = v_1 e_0 + \dots + v_j e_{j-1} + e_j + v_{j+1} e_{j+1} + \dots + v_n e_n$.

↑ the linear part is the orthogonal complement of e_j, then we shift.

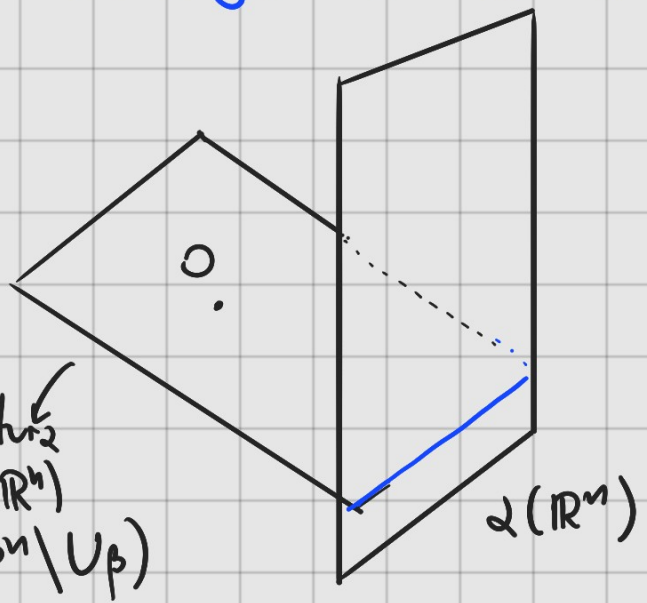


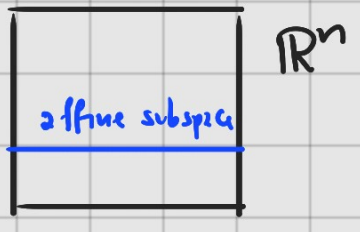
Claim: $\mathbb{RP}^n = \bigcup_{j=0}^{n} U_{\alpha_j}$

Let $L \in \mathbb{RP}^n$. Assume $L \notin U_{\alpha_j}$ $\forall j = 0, \dots, n$.
Then, $L \subset e_j^\perp$ $\forall j = 0, \dots, n$. $\Rightarrow L \subset \{0\}$. $\lightning$

(2) $U_\alpha \cap U_\beta = U_\alpha \setminus (\mathbb{RP}^n \setminus U_\beta)$ all lines that hit $\alpha(\mathbb{R}^n)$ except for some affine subspace.

tutte le rette che non stanno nella giacitura di $\alpha(\mathbb{R}^n)$ né in quella di $\beta(\mathbb{R}^n)$





$$x_\alpha(U_\alpha \cap U_\beta) = x_\alpha(U_\alpha \setminus (\mathbb{RP}^n \setminus U_\beta))$$
$$= \mathbb{R}^n \setminus \{\text{affine subspace}\} \quad \text{is open.}$$

(3) Compute $x_\beta \circ x_\alpha^{-1}$.

$\alpha(v) = Bv + c$
$\beta(v) = Dv + f$

$$W = x_\beta \circ x_\alpha^{-1}(v) \iff x_\beta^{-1}(w) = x_\alpha^{-1}(v)$$
$$\iff \mathbb{R}\beta(w) = \mathbb{R}\alpha(v)$$
$$\iff \exists\, t \in \mathbb{R} : \beta(w) = t\alpha(v)$$

$\Leftrightarrow$ $\underbrace{Dw + f}_{(D,f)\begin{pmatrix} w \\ 1 \end{pmatrix}} = t(Bv+c)$, f is not in the image of D ($\Leftrightarrow$ $\beta(\mathbb{R}^n)$)

$\Rightarrow$ (D,f) is invertible.

$$\begin{pmatrix} w \\ 1 \end{pmatrix} = t(D,f)^{-1}(Bv+c)$$

$$\left\langle \begin{pmatrix} w \\ 1 \end{pmatrix}, e_n \right\rangle = \langle t(D,f)^{-1}(Bv+c), e_n \rangle$$

$$\| \quad 1$$

Solve for t: $t = \langle (D,f)^{-1}(Bv+c), e_n \rangle^{-1}$

$$\begin{pmatrix} w \\ 1 \end{pmatrix} = \frac{1}{\langle (D,f)^{-1}(Bv+c), e_n \rangle} (D,f)^{-1}(Bv+c)$$

We have expressed w in terms of v: w is quotient of certain linear maps $\Rightarrow$ rational function. $\Rightarrow$ $x_\beta \circ x_\alpha^{-1}$ continuous.

PROOF (THM)

Uniqueness: suppose M has such a topology.

$U \subset M$ open $\Leftrightarrow$ $U \cap U_\alpha$ open $\forall \alpha \in A$
$\Leftrightarrow$ $x_\alpha(U \cap U_\alpha) \subset \mathbb{R}^n$ open $\forall \alpha$. (*)

Existence: take (*) as a definition. Check that this defines a topology... $\square$

Addendum 1 to thm: assumptions as in thm 1 + there is a countable subset $A' \subset A$ s.t. $M = \bigcup_{\alpha \in A'} U_\alpha$. Then the topology of M has a countable basis.

PROOF:

$U_\alpha \cong V_\alpha \Rightarrow U_\alpha$ has countable basis $\mathcal{B}_\alpha$.

$\mathcal{B} = \bigcup_{\alpha \in A'} \mathcal{B}_\alpha$ countable basis of M.

□.

- In $\mathbb{RP}^n$: $A' = \{\alpha_0, \dots, \alpha_n\}$ does the job.

Addendum 2 to thm: assumptions as in thm, in addition we assume $\forall p, q \in M \ \exists \alpha \in A$ s.t. $p, q \in U_\alpha$. Then, the topology of M is Hausdorff.

PROOF.

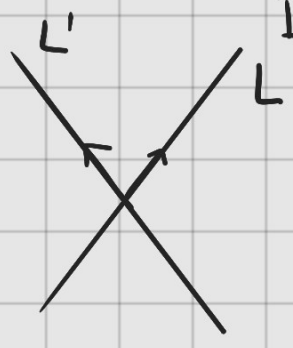
$p \neq q \in M$, let $\alpha \in A$ s.t. $p, q \in U_\alpha \overset{\chi_\alpha}{\cong} V_\alpha$, $\mathbb{R}^n$ Hausdorff.

$\Rightarrow \exists\, V, V' \subset V_\alpha$ open s.t. $\chi_\alpha(p) \in V$, $\chi_\alpha(q) \in V'$, $V \cap V' = \emptyset$.

$\Rightarrow \chi_\alpha^{-1}(V) \ni p$ and $\chi_\alpha^{-1}(V) \cap \chi_\alpha^{-1}(V') = \emptyset$.

$\chi_\alpha^{-1}(V') \ni q$

□.



- In $\mathbb{RP}^n$: let $L, L' \in \mathbb{RP}^n$.

$\exists$ affine-linear embedding

$\alpha : \mathbb{R}^n \to \mathbb{R}^{n+1}$ s.t.

$L \cap \alpha(\mathbb{R}^n) \neq \emptyset$, $L' \cap \alpha(\mathbb{R}^n) \neq \emptyset \Rightarrow L, L' \subset U_\alpha$.

CONCLUSION: $\mathbb{RP}^n$ n-dimensional manifold.

Similarly, $\mathbb{CP}^n = \{L \subset \mathbb{C}^{n+1} \mid \mathbb{C}\text{ 1-dim. vector subspace}\}$

↑ $2n$-dim. manifold.

DEF: two charts $x: U \to V$, $y: U' \to V'$ of a top. manifold are called **C^∞- COMPATIBLE** iff

$$y \circ x^{-1}: x(U \cap U') \longrightarrow y(U \cap U')$$

is a C^∞- diffeomorphism.

DEF: A collection $\mathcal{A} = \{x_2: U_2 \to V_2\}$ of charts of topological manifold is called an **ATLAS** of M iff $M = \bigcup U_2$. $\mathcal{A}$ is called **C^∞- ATLAS** iff any two charts in $\mathcal{A}$ are C^∞- compatible.

RMK: let $\mathcal{A}$ be a C^∞- atlas. If $x, \tilde{x}$ are charts which are C^∞- compatible with all charts in $\mathcal{A}$, then $x, \tilde{x}$ are C^∞- compatible with each other.

$\mathcal{A}_{max} := \{\text{charts } x \mid x \text{ is } C^\infty\text{- compatible with all charts in } \mathcal{A}\}$ is a C^∞- atlas (it is **MAXIMAL C^∞- ATLAS**)

DEF: a maximal C^∞- atlas is called a **DIFFERENTIABLE STRUCTURE** of M.

RMK: each C^∞- atlas is contained in a diff. structure.

DEF: a **DIFFERENTIABLE MANIFOLD** is $(M, \mathcal{A}_{max})$, M topological manifold and $\mathcal{A}_{max}$ differentiable structure.